

$$= \left[(\cos x - \sin x) \frac{d(\cos x + \sin x)}{dx} - (\cos x + \sin x) \frac{d(\cos x - \sin x)}{dx} \right] \times \frac{1}{(\cos x - \sin x)^2}$$

$$= \frac{(\cos x - \sin x) [-\sin x + \cos x] - (\cos x + \sin x) (-\sin x - \cos x)}{(\cos x - \sin x)^2}$$

$$= \frac{(\cos x - \sin x)^2 + (\cos x + \sin x)^2}{(\cos x - \sin x)^2}$$

$$= \frac{\cos^2 x - 2\sin x \cdot \cos x + \sin^2 x + \cos^2 x + 2\sin x \cdot \cos x + \sin^2 x}{(\cos x - \sin x)^2}$$

$$= \frac{2(\sin^2 x + \cos^2 x)}{(\cos x - \sin x)^2}$$

$$= \frac{2}{\cos^2 x + \sin^2 x - 2\sin x \cdot \cos x}$$

$$= \frac{2}{1 - \sin 2x}$$

$$= \frac{2(1 + \sin 2x)}{1 - \sin^2 2x}$$

$$= \frac{2(1 + \sin 2x)}{\cos^2 2x}$$

33) e) $\sqrt{\frac{\sec x + \tan x}{\sec x - \tan x}}$

→ let $y = \sqrt{\frac{\sec x + \tan x}{\sec x - \tan x}}$

$$y = \sqrt{\frac{\sec x + \tan x}{\sec x - \tan x}} \times \frac{\sec x + \tan x}{\sec x + \tan x}$$

$$= \sqrt{\frac{(\sec x + \tan x)^2}{\sec^2 x - \tan^2 x}}$$

$$= \sqrt{\frac{(\sec x + \tan x)^2}{1}}$$

$$y = \sec x + \tan x$$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d(\sec x + \tan x)}{dx}$$

$$= \frac{d \sec x}{dx} + \frac{d \tan x}{dx}$$

$$= \sec x \cdot \tan x + \sec^2 x$$

$$= \sec x (\tan x + \sec x) \quad \downarrow$$

34) f) $\sqrt{\frac{1 - \sin x}{1 + \sin x}} ; 0 \leq x \leq \frac{\pi}{2}$

→ let $y = \sqrt{\frac{1 - \sin x}{1 + \sin x}}$

$$= \sqrt{\frac{1 - \sin x}{1 + \sin x}} \times \frac{1 - \sin x}{1 - \sin x}$$

$$= \sqrt{\frac{(1 - \sin x)^2}{1 - \sin^2 x}}$$

$$= \sqrt{\frac{(1 - \sin x)^2}{\cos^2 x}}$$

$$= \frac{1 - \sin x}{\cos x}$$

$$= \frac{1}{\cos x} - \frac{\sin x}{\cos x}$$

$$= \sec x - \tan x$$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d(\sec x - \tan x)}{dx}$$

$$= \frac{d \sec x}{dx} - \frac{d \tan x}{dx}$$

$$= \sec x \cdot \tan x - \sec^2 x$$

$$= \sec x (\tan x - \sec x)$$

6) Find the derivative of

35 a) $\sin 2x \cdot \cos 3x$

let $y = \sin 2x \cdot \cos 3x$

$$y = \frac{1}{2} \times 2 \sin 2x \cdot \cos 3x$$

$$= \frac{1}{2} [\sin(3x+2x) - \sin(3x-2x)]$$

$$= \frac{1}{2} [\sin 5x - \sin x]$$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{1}{2} \frac{d[\sin 5x - \sin x]}{dx}$$

$$= \frac{1}{2} \left[\frac{d \sin 5x}{dx} - \frac{d \sin x}{dx} \right]$$

~~$$= \frac{1}{2} [\cos 5x - \cos x]$$~~

$$= \frac{1}{2} \left[\frac{d \sin 5x}{dx} \times \frac{dx}{dx} - \frac{d \sin x}{dx} \right]$$

$$= \frac{1}{2} [\cos 5x \times 5 - \cos x]$$

$$= \frac{1}{2} [\sin 5\pi - \cos \pi]$$

36) b) $\cos 4\pi \cdot \sin 6\pi$

→ let $y = \cos 4\pi \cdot \sin 6\pi$

$$y = \frac{1}{2} [2 \sin 6\pi \cdot \cos 4\pi]$$

$$= \frac{1}{2} [\sin(6\pi + 4\pi) + \sin(6\pi - 4\pi)]$$

$$= \frac{1}{2} [\sin 10\pi + \sin 2\pi]$$

∴ differentiating with respect to π .

$$\frac{dy}{d\pi} = \frac{1}{2} \frac{d(\sin 10\pi + \sin 2\pi)}{d\pi}$$

$$= \frac{1}{2} \left[\frac{d \sin 10\pi}{d\pi} + \frac{d \sin 2\pi}{d\pi} \right]$$

$$= \frac{1}{2} \left[\frac{d \sin 10\pi}{d 10\pi} \times \frac{d 10\pi}{d\pi} + \frac{d \sin 2\pi}{d 2\pi} \times \frac{d 2\pi}{d\pi} \right]$$

$$= \frac{1}{2} [\cos 10\pi \times 10 + \cos 2\pi \times 2]$$

$$= \frac{1}{2} [10 \cos 10\pi + 2 \cos 2\pi]$$

$$= \frac{1}{2} (5 \cos 10\pi + \cos 2\pi)$$

$$= 5 \cos 10\pi + \cos 2\pi$$

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37) $\cos 7x \cdot \cos 5x$

$\rightarrow \text{let } y = \cos 7x \cdot \cos 5x$

$$y = \frac{1}{2} [2 \cos 7x \cdot \cos 5x]$$

$$= \frac{1}{2} [\cos(7x+5x) + \cos(7x-5x)]$$

$$= \frac{1}{2} [\cos 12x + \cos 2x]$$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{1}{2} \frac{d[\cos 12x + \cos 2x]}{dx}$$

$$= \frac{1}{2} \left[\frac{d \cos 12x}{dx} + \frac{d \cos 2x}{dx} \right]$$

$$= \frac{1}{2} \left[\frac{d \cos 12x}{d 12x} \times \frac{d 12x}{dx} + \frac{d \cos 2x}{d 2x} \times \frac{d 2x}{dx} \right]$$

$$= \frac{1}{2} [-\sin 12x \times 12 + (-\sin 2x) \times 2]$$

$$= \frac{1}{2} [-12 \sin 12x - 2 \sin 2x]$$

$$= \frac{1}{2} \times -2(6 \sin 12x + \sin 2x)$$

$$= -(6 \sin 12x + \sin 2x)$$

38 d) $\sin 2m\pi \cdot \sin 2n\pi$

→ let $y = \sin 2m\pi \cdot \sin 2n\pi$

$$y = \frac{1}{2} [2 \sin 2m\pi \cdot \sin 2n\pi]$$

$$= \frac{1}{2} [\cos(2m\pi - 2n\pi) - \cos(2m\pi + 2n\pi)]$$

$$= \frac{1}{2} [\cos 2\pi(m-n) - \cos 2\pi(m+n)]$$

differentiating with respect to π .

$$\frac{dy}{d\pi} = \frac{1}{2} \frac{d[\cos 2\pi(m-n) - \cos 2\pi(m+n)]}{d\pi}$$

$$= \frac{1}{2} \left[\frac{d \cos 2\pi(m-n)}{d\pi} - \frac{d \cos 2\pi(m+n)}{d\pi} \right]$$

$$= \frac{1}{2} \left[\frac{d \cos 2\pi(m-n)}{d 2\pi(m-n)} \times \frac{d 2\pi(m-n)}{d\pi} - \frac{d \cos 2\pi(m+n)}{d 2\pi(m+n)} \times \frac{d 2\pi(m+n)}{d\pi} \right]$$

$$= \frac{1}{2} [-\sin 2\pi(m-n) \times 2(m-n) - \sin 2\pi(m+n) \times 2(m+n)]$$

$$= -[(\sin 2\pi(m-n) \times (m-n) + \sin 2\pi(m+n) \times (m+n))]$$

$$= -[(m-n) \cdot \sin 2\pi(m-n) + (m+n) \sin 2\pi(m+n)]$$

7) find $\frac{dy}{dx}$ of the following.

49 a) $x+y = \cos y$

differentiating with respect to x .

$$\frac{d(x+y)}{dx} = \frac{d \cos y}{dx}$$

$$\text{or, } \frac{dx}{dx} + \frac{dy}{dx} = \frac{d \cos y}{dy} \times \frac{dy}{dx}$$

$$\text{or, } 1 + \frac{dy}{dx} = -\sin y \times \frac{dy}{dx}$$

$$\text{or, } \frac{dy}{dx} (1 + \sin y) = -1$$

$$\frac{dy}{dx} = \frac{-1}{1 + \sin y}$$

$$\therefore \frac{dy}{dx} = \frac{-1}{1 + \sin y} \quad \checkmark$$

40 b) $\sin^2 x + 2 \cos y + xy = 0$

differentiating with respect to x .

$$\frac{d \sin^2 x}{dx} + \frac{d 2 \cos y}{dx} + \frac{d xy}{dx} = 0$$

$$\text{or, } \frac{d \sin^2 x}{d \sin x} \times \frac{dx}{dx} + 2 \frac{d \cos y}{dy} \times \frac{dy}{dx} + \left(y \frac{dx}{dx} + x \frac{dy}{dx} \right) = 0$$

$$\text{or, } 2 \sin x \times 1 + 2(-\sin y) \times \frac{dy}{dx} + y + x \cdot \frac{dy}{dx} = 0$$

$$\text{or, } \frac{dy}{dx} (x - 2 \sin y) = -y - 2 \sin x$$

$$\text{or, } \frac{dy}{dx} = \frac{-(y + 2 \sin x)}{(x - 2 \sin y)}$$

$$\therefore \frac{dy}{dx} = \frac{y + 2 \sin x}{2 \sin y + x} \quad \checkmark$$

41) c) $x - y = \sec xy$

→ differentiating with respect to x .

$$\frac{dx}{dx} - \frac{dy}{dx} = \frac{d \sec xy}{dx}$$

$$\text{or, } 1 - \frac{dy}{dx} = \frac{d \sec xy}{dxy} \times \frac{dxy}{dx}$$

$$\text{or, } 1 - \frac{dy}{dx} = \sec xy \cdot \tan xy \times \left[x \frac{dy}{dx} + y \cdot \frac{dx}{dx} \right]$$

$$\text{or, } 1 - \frac{dy}{dx} = \sec xy \cdot \tan xy \left[x \cdot \frac{dy}{dx} + y \right]$$

$$\text{or, } 1 - \frac{dy}{dx} = x \sec xy \cdot \tan xy \cdot \frac{dy}{dx} + y \sec xy \cdot \tan xy$$

$$\text{or, } \frac{dy}{dx} [x \sec xy \cdot \tan xy + 1] = 1 - y \sec xy \cdot \tan xy$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - y \sec xy \cdot \tan xy}{x \sec xy \cdot \tan xy + 1}$$

42) d) $x^2 + y^2 = \sin xy$

→ differentiating with respect to x .

$$\frac{dx^2}{dx} + \frac{dy^2}{dx} = \frac{d \sin xy}{dx}$$

$$\text{or, } 2x + \frac{dy^2}{dy} \times \frac{dy}{dx} = \frac{d \sin xy}{dxy} \times \frac{dxy}{dx}$$

$$\text{or, } 2x + 2y \times \frac{dy}{dx} = \cos xy \times \left[x \frac{dy}{dx} + y \right]$$

$$\text{or, } 2x + 2y \cdot \frac{dy}{dx} = x \cos xy \times \frac{dy}{dx} + y \cos xy$$

$$\text{or, } \frac{dy}{dx} [2y - x \cos xy] = y \cos xy - 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \cos xy - 2x}{2y - x \cos xy}$$

43 e) $xy = \tan(x^2 + y^2)$

→ differentiating with respect to x .

$$\frac{dxy}{dx} = \frac{d(\tan(x^2 + y^2))}{dx}$$

$$\text{or, } x \frac{dy}{dx} + y \frac{dx}{dx} = \frac{d(\tan(x^2 + y^2))}{d(x^2 + y^2)} \times \frac{d(x^2 + y^2)}{dx}$$

$$\text{or, } x \cdot \frac{dy}{dx} + y = \sec^2(x^2 + y^2) \left[\frac{dx^2}{dx} + \frac{dy^2}{dy} \times \frac{dy}{dx} \right]$$

$$\text{or, } x \cdot \frac{dy}{dx} + y = \sec^2(x^2 + y^2) \left[2x + 2y \times \frac{dy}{dx} \right]$$

$$\text{or, } x \cdot \frac{dy}{dx} + y = 2x \sec^2(x^2 + y^2) + 2y \sec^2(x^2 + y^2) \times \frac{dy}{dx}$$

$$\text{or, } \frac{dy}{dx} [x - 2y \sec^2(x^2 + y^2)] = 2x \sec^2(x^2 + y^2) - y$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x \sec^2(x^2 + y^2) - y}{x - 2y \sec^2(x^2 + y^2)}$$

44 f) $x^2 y = \sec xy^2$

→ differentiating with respect to x .

$$\frac{d(x^2 y)}{dx} = \frac{d(\sec xy^2)}{dx}$$

$$\text{or, } y \frac{dx^2}{dx} + x^2 \frac{dy}{dx} = \frac{d(\sec xy^2)}{d(xy^2)} \times \frac{d(xy^2)}{dx}$$

$$\text{or, } 2xy + x^2 \frac{dy}{dx} = \sec xy^2 \cdot \tan xy^2 \times \left[x \frac{dy^2}{dx} + y^2 \frac{dx}{dx} \right]$$

$$\text{or, } 2xy + x^2 \frac{dy}{dx} = x \sec xy^2 \cdot \tan xy^2 \times 2y \times \frac{dy}{dx}$$

$$+ y^2 \sec xy^2 \cdot \tan xy^2$$

$$\text{or, } \frac{dy}{dx} [x^2 - 2xy \cdot \sec xy^2 \cdot \tan xy^2] = y^2 \sec xy^2 \cdot \tan xy^2 - 2xy$$

$$\frac{dy}{dx} = \frac{y^2 \sec xy^2 \cdot \tan xy^2 - 2xy}{x^2 - 2xy \cdot \sec xy^2 \cdot \tan xy^2}$$

8. Find $\frac{dy}{dx}$, when

(45) a) $y = \sin \theta$, $\theta = 5x^2 - 6x + 2$
 $\rightarrow y = \sin \theta \rightarrow (i)$
 $\theta = 5x^2 - 6x + 2 \rightarrow (ii)$

differentiating (i) with respect to θ .

$$\frac{dy}{d\theta} = \frac{d \sin \theta}{d\theta}$$

$$\frac{dy}{d\theta} = \cos \theta$$

differentiating (ii) with respect to x .

$$\frac{d\theta}{dx} = \frac{d(5x^2 - 6x + 2)}{dx}$$

$$= \frac{d5x^2}{dx} - \frac{d6x}{dx} + \frac{d2}{dx}$$

$$= 10x - 6$$

Now,

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \cos \theta \times (10x - 6)$$

$$= (10x - 6) \cos(5x^2 - 6x + 2)$$

46b) $x = a \cos^2 \theta$, $y = b \sin^2 \theta$

$\rightarrow x = a \cos^2 \theta \rightarrow (i)$

$y = b \sin^2 \theta \rightarrow (ii)$

differentiating (i) with respect to θ

$$\frac{dx}{d\theta} = a \frac{d \cos^2 \theta}{d\theta}$$

$$= a \frac{d \cos^2 \theta}{d \cos \theta} \times \frac{d \cos \theta}{d\theta}$$

$$= a 2 \cos \theta \times (-\sin \theta)$$

$$= -a \sin 2\theta$$

differentiating (ii) with respect to θ

$$\frac{dy}{d\theta} = b \frac{d \sin^2 \theta}{d\theta}$$

$$= b \frac{d \sin^2 \theta}{d \sin \theta} \times \frac{d \sin \theta}{d\theta}$$

$$= b 2 \sin \theta \times \cos \theta$$

$$= b \sin 2\theta$$

Now,

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= b \sin 2\theta \times \frac{1}{-a \sin 2\theta}$$

$$= -\frac{b}{a}$$

$$= -\frac{b}{a}$$

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(17) c) $x = \tan t$, $y = \sin t \cdot \cos t$

$\rightarrow x = \tan t \rightarrow (i)$

$y = \sin t \cdot \cos t \rightarrow (ii)$

differentiating (i) with respect to ~~x~~ t .

$$\frac{dx}{dt} = \frac{d \tan t}{dt}$$

$$= \sec^2 t$$

differentiating (ii) with respect to t

$$\frac{dy}{dt} = \frac{d(\sin t \cdot \cos t)}{dt}$$

$$= \sin t \frac{d \cos t}{dt} + \cos t \cdot \frac{d \sin t}{dt}$$

$$= \sin t (-\sin t) + \cos t (\cos t)$$

$$= -\sin^2 t + \cos^2 t$$

Now,

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= -\sin^2 t + \cos^2 t \times \frac{1}{\sec^2 t}$$

$$= (\cos^2 t - \sin^2 t) \cos^2 t$$

$$\therefore \frac{dy}{dx} = \cos^2 t (\cos^2 t - \sin^2 t) \quad \text{✓}$$

48) $x = a \sin t$, $y = a \cos t$

$\rightarrow x = a \sin t \rightarrow (i)$

$y = a \cos t \rightarrow (ii)$

differentiating (i) with respect to t .

$$\frac{dx}{dy} \cdot \frac{dy}{dt} = \frac{a \sin t}{dt}$$
$$= a \cos t$$

differentiating (ii) with respect to t .

$$\frac{dy}{dt} = \frac{a \cos t}{dt}$$
$$= a(-\sin t)$$
$$= -a \sin t$$

Now,

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= -\sin t \times \frac{1}{a \cos t}$$

$$= -\frac{\sin t}{\cos t}$$

$$= -\tan t$$

$$\therefore \frac{dy}{dx} = -\tan t$$

Q9) $x = a(t + \sin t)$, $y = a(1 - \cos t)$

$\rightarrow x = a(t + \sin t) \rightarrow (i)$

$y = a(1 - \cos t) \rightarrow (ii)$

differentiating (i) with respect to t .

$$\frac{dx}{dt} = a \frac{d(t + \sin t)}{dt}$$
$$= a \left[\frac{dt}{dt} + \frac{d \sin t}{dt} \right]$$

$$= a[1 + \cos t]$$

differentiating (ii) with respect to t .

$$\frac{dy}{dt} = a \frac{d(1 - \cos t)}{dt}$$
$$= a \left[\frac{d1}{dt} - \frac{d \cos t}{dt} \right]$$

$$= a[-(-\sin t)]$$

$$= a \sin t$$

Now,

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= \cancel{a} \sin t \times \frac{1}{\cancel{a}(1 + \cos t)}$$

$$= \frac{\sin t (1 - \cos t)}{\sin^2 t}$$

$$= \operatorname{cosec} t - \cot t$$

9) Differentiate:

(50) a) $\cos x$ with respect to $\sin x$.

→ Soln,

$$\begin{aligned} & \frac{d \cos x}{d \sin x} \\ &= \frac{\frac{d \cos x}{dx}}{\frac{d \sin x}{dx}} \\ &= \frac{-\sin x}{\cos x} \\ &= -\tan x \quad \text{Ans.} \end{aligned}$$

(51) b) $\sec x$ with respect to $\tan x$.

→ Soln,

$$\begin{aligned} & \frac{d \sec x}{d \tan x} \\ &= \frac{\frac{d \sec x}{dx}}{\frac{d \tan x}{dx}} \\ &= \frac{\sec x \cdot \tan x}{\sec^2 x} \\ &= \tan x \times \cos x \\ &= \frac{\sin x}{\cancel{\cos x}} \times \cos x \\ &= \sin x \quad \text{Ans.} \end{aligned}$$

Q5) $\tan \theta$ with respect to $\cot \theta$
→ Soln,

$$\begin{aligned} & \frac{d \tan \theta}{d \cot \theta} \\ &= \frac{\frac{d \tan \theta}{d \theta}}{\frac{d \cot \theta}{d \theta}} \\ &= \frac{\sec^2 \theta}{-\operatorname{cosec}^2 \theta} \\ &= -\frac{1}{\cos^2 \theta} \times \sin^2 \theta \\ &= -\tan^2 \theta \end{aligned}$$

10) Differentiate:

Q3) a) $\tan \theta^\circ$
 $= \tan\left(\frac{\pi}{180}\right)^\circ$

differentiating with respect to θ .

$$\begin{aligned} & \frac{d \tan\left(\frac{\pi \theta}{180}\right)}{d \theta} \\ &= \frac{d \tan\left(\frac{\pi \theta}{180}\right)}{d\left(\frac{\pi \theta}{180}\right)} \times \frac{d\left(\frac{\pi \theta}{180}\right)}{d \theta} \\ &= \sec^2 \frac{\pi \theta}{180} \times \frac{\pi}{180} \\ &= \frac{\pi}{180} \times \sec^2 \frac{\pi \theta}{180} \end{aligned}$$

54) b) $\sin n^\circ$

$$= \sin \frac{n^\circ}{200}$$

differentiating with respect to n .

$$\frac{d \sin \left(\frac{n^\circ}{200} \right)}{dn}$$

$$= \frac{d \sin \left(\frac{n^\circ}{200} \right)}{d \left(\frac{n^\circ}{200} \right)} \times \frac{d \left(\frac{n^\circ}{200} \right)}{dn}$$

$$= \cos \frac{n^\circ}{200} \times \frac{n^\circ}{200}$$

$$= \frac{n^\circ}{200} \times \cos \frac{n^\circ}{200} \quad \text{u.}$$

55) c) $n \cos n^\circ$

$$= n \cos \frac{n^\circ}{180}$$

differentiating with respect to n .

$$= \frac{d \left[n \cos \frac{n^\circ}{180} \right]}{dn}$$

$$= n \frac{d \cos \frac{n^\circ}{180}}{dn} + \cos \frac{n^\circ}{180} \times \frac{dn}{dn}$$

$$= n \times -\sin \left(\frac{n^\circ}{180} \right) \times \frac{n^\circ}{180} + \cos \frac{n^\circ}{180}$$

$$= \cos \frac{n^\circ}{180} - \frac{n^\circ}{180} \times \sin \left(\frac{n^\circ}{180} \right) \quad \text{u.}$$

Q 56 i) $\sqrt{\operatorname{cosec} x}$
 let $f(x) = \sqrt{\operatorname{cosec} x}$
 $f(x+h) = \sqrt{\operatorname{cosec}(x+h)}$

Where h is small change in x .

By definition of derivative,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{\operatorname{cosec}(x+h)} - \sqrt{\operatorname{cosec} x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{\operatorname{cosec}(x+h)} - \sqrt{\operatorname{cosec} x} \times \sqrt{\operatorname{cosec}(x+h)} + \sqrt{\operatorname{cosec} x}}{h(\sqrt{\operatorname{cosec}(x+h)} + \sqrt{\operatorname{cosec} x})}$$

$$= \lim_{h \rightarrow 0} \frac{\operatorname{cosec}(x+h) - \operatorname{cosec} x}{h(\sqrt{\operatorname{cosec}(x+h)} + \sqrt{\operatorname{cosec} x})}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h(\sqrt{\operatorname{cosec}(x+h)} + \sqrt{\operatorname{cosec} x})} \times \left[\frac{1}{\sin(x+h)} - \frac{1}{\sin x} \right]$$

$$= \frac{1}{2\sqrt{\operatorname{cosec} x}} \times \lim_{h \rightarrow 0} \frac{\sin x - \sin(x+h)}{\sin x \cdot \sin(x+h)} \times \frac{1}{h}$$

$$= \frac{1}{2\sqrt{\operatorname{cosec} x}} \times \frac{1}{\sin^2 x} \times \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{x - (x+h)}{2}\right)}{2}$$

$$\cdot \cos\left(\frac{x+x+h}{2}\right) \times \frac{1}{h}$$

$$= \frac{1}{2\sqrt{\operatorname{cosec} x} \times \sin^2 x} \times \lim_{h \rightarrow 0} \frac{(-h) \sin h}{\frac{h}{2} \times \frac{1}{2} \times \cos(2x+h)}$$

$$= \frac{1}{2\sqrt{\operatorname{cosec} x} \times \sin^2 x} \times \lim_{h \rightarrow 0} \frac{(-h) \sin h}{\frac{h}{2} \times \frac{1}{2} \times \cos(2x+h)}$$

$$= \frac{1}{2\sqrt{\operatorname{cosec} x} \times \sin^2 x} \times -1 \times \cos x$$

$$= \frac{-\cot x \cdot \sqrt{\operatorname{cosec} x}}{2} \quad \because \sin x = \frac{1}{\operatorname{cosec} x} = \frac{1}{\sqrt{\operatorname{cosec}^2 x}}$$